Simone MACHETTI

PERSONAL DATA

PLACE AND DATE OF BIRTH	Turin, Italy | 13 September 1993
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PROFILE

Researcher with 7+ years of experience in low-power heterogeneous architectures, with expertise in RISC-V platforms, accelerator integration (CGRA, IMC, NMC), and GPU design for TinyAI. Experienced in both silicon tapeouts (65 nm, 16 nm) and FPGA prototyping. Author of 10+ publications with more than 130 citations at top conferences and journals (ICCAD, ISVLSI, IEEE Micro, TBioCAS). Co-organizer of international workshops, teaching assistant in ASIC/FPGA design, and active contributor to open-source frameworks (FEMU, X-HEEP, e-GPU).

EDUCATION AND WORK

2025 - now | **Research Member at SwissChips, Lausanne**

I am part of the SwissChips project to advance open-source and energy-efficient hardware design in Switzerland.

2020 - now | **Senior Researcher at X-HEEP Platform, Lausanne**

This university project focuses on the development of open-source, configurable, and extendible RISC-V hardware. It is built around the eXtendible Heterogeneous Energy Efficient Platform (X-HEEP), a configurable and extendible RISC-V host designed to support the exploration of ultra-low-power edge accelerators.

2020 - now | **PhD Researcher in Electronic Engineering at the Embedded Systems Laboratory (ESL), EPFL, Prof. David Atienza**

THESIS TITLE: Open-Source and Configurable RISC-V Platforms for Exploring TinyAI Heterogeneous Systems.

MAIN CONTRIBUTIONS:

○ **FPGA EMUlation (FEMU)**, an open-source and configurable emulation framework for prototyping and evaluating TinyAI heterogeneous systems.

○ **eXtendible Heterogeneous Energy Efficient Platform (X-HEEP)**, an open-source, configurable, and extendible RISC-V platform for supporting the exploration of TinyAI accelerators.

○ **embedded GPU (e-GPU)**, an open-source and configurable RISC-V platform for exploring the feasibility and trade-offs of utilizing GPUs in TinyAI scenarios.

TAPEOUTS:

○ **HEEPocrates - 65 nm TSMC**, which combines the X-HEEP host with a coarse-grained reconfigurable array (CGRA) and in-memory computing (IMC) accelerators.

○ **HEEPatia - 16 nm TSMC**, which combines the X-HEEP host with a coarse-grained reconfigurable array (CGRA) and near-memory computing (NMC) accelerators, and a co-processor for posit arithmetics.

WORKSHOPS:

○ **DATE24 Conference**, I co-organized a workshop on RISC-V open-source hardware and software, where I talked about my research activities.

COMPETITIONS:

○ **AMD Open Hardware**, which focused on the design of an emulation platform for exploring TinyAI heterogeneous systems.

HONORS:

○ **Best Project Awards**, issued by SMARTHEP Edge Machine Learning School at CERN, Geneva.

2020 - now | **Teaching Assistant, EPFL**

COURSES:

○ **Lab in advanced VLSI design**, which focused on the complete ASIC design flow, from RTL to GDS, with Prof. Andreas Burg.

○ **Digital systems design**, which focused on the entry-level FPGA design flow, using Xilinx FPGAs, with Prof. Andreas Burg.

○ **Lab on hardware-software digital systems co-design**, which focused on the design of complex hw/sw architectures on Xilinx FPGAs, with Prof. David Atienza.

2019 - 2020 | **Research Intern at the Embedded Systems Laboratory (ESL)**, EPFL

The Internship focused on the implementation and optimization of convolutional neural networks (CNN) on ultra-low-power heterogeneous multi-core architectures. This project was in collaboration with Nespresso.

2019 - 2020 | **External Researcher at Nestlé Nespresso**, Lausanne

I collaborated with Nespresso during my Internship at the Embedded Systems Laboratory (ESL) of EPFL. The work focused on the implementation and optimization of convolutional neural networks (CNN) on ultra-low-power heterogeneous multi-core architectures.

2018 - 2019 | **Firmware Engineer at SPEA - Automatic Test Equipment**, Turin

The work focused on the development of low-level firmware to implement different functionalities of automatic test equipment (ATE). I also trained to use the complete laboratory equipment, including oscilloscopes, multimeters, wave generators, soldering irons, etc.

2016 - 2018 | **Master Degree in Computer Engineering** (courses fully taught in English), Politecnico di Torino, Turin

MAJOR: **Digital Design**

AVERAGE SCORE: **29.5/30**

FINAL EVALUATION: **110 Summa Cum Laude**

THESIS WORK:

ASIP Design for Motion Estimation in Video Compression Algorithms.

The project focused on the design of an ASIP microprocessor core optimized for video compression applications. ASIP Designer by Synopsys was used to describe, compile, and simulate the processor.

RTL PROJECTS:

◦ **CPU RTL Design**, which focused on the design of a DLX microprocessor core with 5 pipeline stages.

◦ **Filter RTL Design**, which focused on the design of a finite-impulse response low-pass filter (FIR) at different abstraction levels.

◦ **Multiplier RTL Design**, which focused on the design of a modified booth encoding (MBE) multiplier.

◦ **Sniffer RTL Design**, which focused on the design of a bus sniffer used to spy on the instruction bus of an MCU and activate hardware Trojans.

◦ **Interface RTL Design**, which focused on the design of a smart and efficient communication protocol between an MCU and an FPGA.

2013 - 2016 | **Bachelor Degree in Computer Engineering** (courses fully taught in English), Politecnico di Torino, Turin

PROJECTS:

◦ **Smart Bus Station**, which focused on the design of a smart bus station to support and help visually impaired people use the public transport system.

PUBLICATIONS

Papers

◦ FEMU: An Open-Source and Configurable Emulation Framework for Prototyping TinyAI Heterogeneous Systems, IEEE ICCAD 2025

◦ X-HEEP: An Open-Source, Configurable and Extendible RISC-V Platform for TinyAI Applications, IEEE ISVLSI 2025

◦ e-GPU: An Open-Source and Configurable RISC-V Graphic Processing Unit for TinyAI Applications, Preprint 2025

◦ ACE: Automated Optimization Towards Iterative Classification in Edge Health Monitors, IEEE TBioCAS 2025

◦ X-HEEP: An Open-Source, Configurable and Extendible RISC-V Microcontroller for the Exploration of Ultra-Low-Power Edge Accelerators, Preprint 2024

◦ X-HEEP: An Open-Source, Configurable and Extendible RISC-V Microcontroller, ACM CF 2023

- ○ A hardware/software co-design vision for deep learning at the edge, IEEE Micro 2022
- ○ Modular Design and Optimization of Biomedical Applications for Ultra-Low Power Heterogeneous Platforms, IEEE TCAD/ESWeek 2020
- ○ Defeating hardware Trojan in microprocessor cores through software obfuscation, IEEE LATS 2018

Posters

- ○ An Open-Source and Configurable RISC-V CPU/GPU Accelerated Processing Unit for Ultra-Low-Power Wearable Devices, SMARTHEP CERN 2025
- ○ FEMU: An Open-Source RISC-V Emulation Platform for the Exploration of Accelerator-based Edge Applications, DATE 2024
- ○ X-HEEP: An Open-Source, Configurable and Extendible RISC-V Microcontroller for the Exploration of Ultra-Low-Power Edge Accelerators, EcoCloud Event 2023

Reports

- ○ HEEPocrates: An Ultra-Low-Power RISC-V Microcontroller for Edge-Computing Healthcare Applications, Europractice Activity Report 2023

Thesis

- ○ ASIP Design for Motion Estimation in Video Compression Algorithms, Politecnico di Torino, Italy 2018

Computer skills

Hardware	SystemVerilog, Verilog, VHDL
EDA Tools	Modelsim, Design Compiler, Innovus, PrimePower, PrimeTime, Vivado
Scripting	Python, TCL, Bash, Git
Software	C, C++, RISCV Assembly
Domains	MCU/SoC Design, Accelerators (CGRA, IMC, NMC), GPU Architecture, FPGA Prototyping, ASIC Implementation

Languages

Italian	Native language
English	Professional working proficiency | IELTS certification

Sports

Aikido	Japanese self-defence martial art. I am currently Black Belt 4° Dan and in 2015 I obtained the qualification of Fuku Shidon (instructor).

Other certifications

Delpho Didattica Informatica	Computer Hardware Technician
Divers Alert Network	Basic Life Support (BLS)
ICDL Certification	European Computer Driving License
British Council	IELTS Certification